Public Utilities – Hyderabad Smart City Knowledge Chatbot (RAG)

The **Public Utilities** module of the Hyderabad Smart City Knowledge Chatbot is designed to provide comprehensive, accurate, and up-to-date information about all essential public utility services and infrastructure across Hyderabad. Powered by **Retrieval-Augmented Generation (RAG)**, the chatbot retrieves relevant information from official government documents, policies, engineering manuals, service guidelines, annual reports, government orders, departmental notifications, citizen charters, infrastructure reports, and Smart City publications before generating responses. This ensures that every answer is reliable, document-backed, and based on the latest official information. The chatbot serves as a centralized knowledge platform for citizens, government officials, researchers, students, businesses, developers, and visitors who need trusted information about Hyderabad's public utility ecosystem.

The knowledge base covers every major public utility managed by government organizations such as the Greater Hyderabad Municipal Corporation (GHMC), Hyderabad Metropolitan Water Supply and Sewerage Board (HMWSSB), Hyderabad Metropolitan Development Authority (HMDA), Telangana State Southern Power Distribution Company Limited (TSSPDCL), Telangana Municipal Administration and Urban Development Department (MA&UD), Telangana State Pollution Control Board (TSPCB), Telangana State Disaster Response and Fire Services Department, Smart Cities Mission, and other authorized agencies. The chatbot can answer questions related to drinking water supply systems, water sources, reservoirs, water treatment plants, pumping stations, distribution networks, underground reservoirs, overhead service reservoirs, domestic, commercial, and industrial water connections, application procedures, required documents, water tariffs, billing systems, smart water meters, online bill payment, water quality standards, laboratory testing, pipeline maintenance, leakage reporting, emergency water tanker services, rainwater harvesting regulations, groundwater recharge, borewell permissions, water conservation initiatives, and drought management.

The chatbot also provides detailed information about Hyderabad's sewerage and wastewater management infrastructure, including underground sewer networks, sewer connections, sewage treatment plants (STPs), wastewater treatment processes, recycled water reuse, drainage systems, manhole maintenance, septic tank regulations, sewer blockage reporting, stormwater drains, flood mitigation infrastructure, the Strategic Nala Development Programme (SNDP), drain desilting schedules, urban flood management, monsoon preparedness, drainage maintenance, and pollution prevention measures. Users can obtain information about electricity distribution services, domestic and commercial power connections, electricity billing, smart meters, transformer maintenance, scheduled power outages, emergency restoration procedures, underground cabling, rooftop solar installations, net metering, renewable energy initiatives, energy conservation programmes, electrical safety standards, and public street lighting systems, including LED street lights, smart lighting projects, high-mast lighting, fault reporting, and maintenance schedules.

The Public Utilities module also includes information about gas supply services, piped natural gas (PNG), LPG distribution, domestic and commercial gas connections, gas safety regulations, emergency response procedures, and government subsidy programmes. It provides complete details about Hyderabad's public infrastructure, including roads, footpaths, bridges, flyovers, public parks, urban forests, walking tracks, open gyms, playgrounds, community halls, public toilets, bus shelters, smart poles, lakefront development projects, heritage infrastructure, public buildings, and city beautification initiatives. The chatbot further supports questions about digital infrastructure and smart city technologies such as the Integrated Command and Control Centre (ICCC), IoT-enabled utility monitoring, GIS-based asset mapping, smart sensors, utility automation, public Wi-Fi infrastructure, digital governance platforms, mobile applications, online citizen services, digital complaint systems, utility dashboards, and open government data initiatives.

In addition, the chatbot provides information about emergency public utility services, including fire services, disaster response infrastructure, flood preparedness, emergency utility restoration, safety guidelines, public emergency contacts, and incident reporting procedures. Users can also ask questions about preventive maintenance programmes, infrastructure inspections, utility asset management, repair schedules, modernization projects, environmental sustainability initiatives, climate resilience strategies, renewable energy adoption, water conservation programmes, green infrastructure, and sustainable urban development policies. The chatbot assists citizens with applying for utility services, registering complaints, tracking service requests, accessing online government portals, understanding utility regulations, obtaining departmental contact details, learning about service timelines, and finding answers to frequently asked questions.

Using semantic search and document retrieval, the RAG system identifies the most relevant information from official Hyderabad government documents and generates context-aware, accurate, and trustworthy responses. Whether users need information about utility regulations, service procedures, infrastructure projects, maintenance activities, emergency services, smart city initiatives, citizen services, environmental programmes, or departmental responsibilities, the chatbot provides detailed and verified answers based on authoritative government sources, making it a comprehensive digital knowledge assistant for Hyderabad's public utility services.